BRITISH COAL & BTU-LINKED DERIVATIVES

An investment framework for energy-content-adjusted coal pricing, risk transfer and conditional domestic supply optionality.

01

BTU NORMALIZATION

Convert tonne prices to energy cost, while preserving grade and delivery adjustments.

02

DEMAND REALISM

Aggregate growth can support some industry, but does not guarantee coal-demand growth.

03

DERIVATIVES FIRST

Use liquid international benchmarks first; treat a UK index as a gated future option.

MARKET-SCOPE NOTE

UK coal demand fell 56% to 948,000 tonnes in 2025; domestic production was 120,000 tonnes, and coal-fired power demand was zero. This report treats a British domestic Btu index as conditional, and prioritises liquid seaborne benchmarks for immediate risk-management use.

01 / INVESTMENT COMMITTEE SUMMARY

British coal & Btu-linked derivatives

A conditional investment framework for energy-content-adjusted coal pricing, derivatives and domestic supply optionality. The case is materially more constrained than the North Sea gas opportunity.

0.948 Mt
UK coal demand in 2025 [1]

0.120 Mt
UK mined production in 2025 [1]

23.81 MMBtu/t
Approx. energy in 6,000 kcal/kg NAR coal

INVESTMENT VERDICT

Launch a coal risk-management capability first; do not underwrite new domestic thermal-coal mining solely on a thesis of rising aggregate demand. UK coal demand has structurally contracted, coal-fired power generation has ended, and the government has announced an intention to restrict new mine licensing. The scalable near-term derivatives proposition is more likely to use established seaborne benchmarks, with UK-specific Btu-linked products added only when an observable local market exists.

The core opportunity

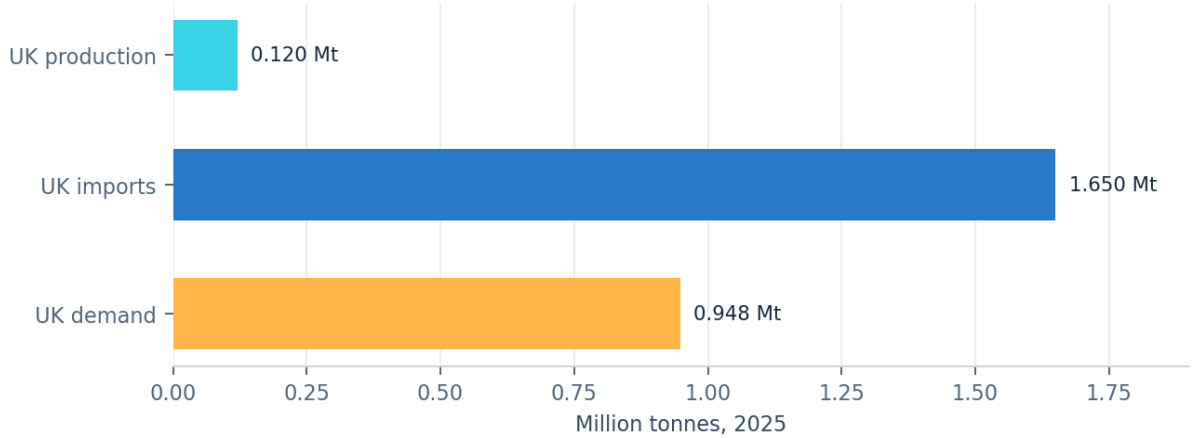
- Monetise price risk, not a directional coal forecast. Provide futures, options, swaps and basis hedges for industrial users, importers, traders and the small population of domestic producers.
- Normalize price by energy content. Express the economic price in USD or GBP per million British thermal units (MMBtu), while retaining the physical contract terms and quality adjustments of the underlying coal.
- Keep the demand thesis honest. Aggregate demand growth may lift activity in steel, cement, chemicals and other industrial sectors, but does not automatically increase UK coal demand; electrification, steel-process change and climate policy can outweigh the macro effect.
- Separate tradable now from speculative later. API2 Rotterdam and Newcastle futures already exist. A new British domestic-coal index would require verified transactions, quality specifications, sufficient reporting and an index-governance framework.

Decision sought: approve a staged feasibility programme for coal derivatives and Btu-based pricing analytics, subject to market-liquidity, conduct, environmental-policy and credit-risk gates. This is a strategic discussion document, not an official Barclays prospectus or investment offer.

02 / MARKET BACKDROP

The UK coal market is small and contracting

The latest official annual data provide a hard constraint on any investment case premised on an immediate domestic coal-demand revival.



UK coal supply and use, 2025. The categories are shown side-by-side to convey scale; production, imports and demand are different flow measures and should not be summed. Source: DESNZ, DUKES 2026. [1]

Indicator	2025 outcome	Investment implication
Total coal demand	948,000 tonnes, down 56% year on year	Demand expansion must be proven by sector; it should not be assumed from GDP growth.
Domestic production	120,000 tonnes, up 12% year on year but around 99% below 2015 levels	Existing local volumes are too small to support a deep, standalone futures market.
Imports	1.65 million tonnes, down 6.3%	The UK market is mainly import-supplied; international benchmark exposure is commercially relevant.
Power-sector demand	Zero coal consumption for electricity generation in 2025	Do not build the thesis around a return of coal-fired generation.
Domestic coal mix	All reported 2025 production was deep-mined, almost exclusively anthracite	Domestic coal is not interchangeable with a 6,000 kcal/kg seaborne thermal-coal benchmark.
Industry	Iron and steel used 207,000 tonnes of coal, down 65% year on year	Potential demand pockets are industrial and grade-specific, not broad-based.

POLICY CONSTRAINT

The government announced in November 2024 that it intended to legislate to restrict future licences for new coal mines. A 2026 tracker reported that no licences for new sites had been issued since the 2024 election, while legislation remained a stated intention. Treat new mine development as a major permitting and policy risk, not a base-case assumption. [2][3]

03 / ENERGY-CONTENT NORMALIZATION

Use MMBtu as the common economic unit

A Btu-based lens makes coal grades easier to compare economically, but it does not erase differences in quality, delivery point, carbon content or contractual specification.

Unit definitions and conversion

Item	Working definition	Use in the model
Btu	British thermal unit: a unit of heat energy	Measures energy content; not itself a commodity price.
MMBtu	1 million Btu	Preferred reporting unit for energy-adjusted price and cost.
API2 reference grade	6,000 kcal/kg NAR thermal coal delivered CIF ARA	Standard seaborne benchmark basis; not a specification for British anthracite. [4][5]
Energy equivalence	6,000 kcal/kg ≈ 10,800 Btu/lb ≈ 23.81 MMBtu per metric tonne	Approximation for conversion; actual cargo heat value varies.

PRICE CONVERSION

USD/MMBtu = USD per metric tonne ÷ MMBtu per tonne.
At the 6,000 kcal/kg NAR reference basis: USD/MMBtu ≈ USD/t ÷ 23.81. For an actual cargo, replace 23.81 with its measured net calorific value converted to MMBtu/tonne, then separately adjust for quality, freight, handling, taxes and emissions costs where applicable.

Illustrative benchmark conversion - not live market quotations

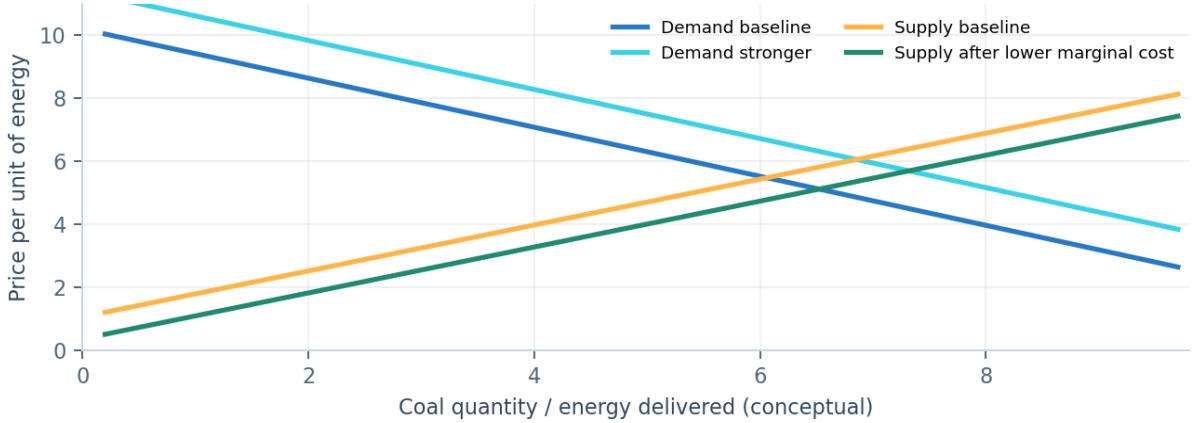
Illustrative coal price	Equivalent at 23.81 MMBtu/t	Interpretation
$75/t	$3.15/MMBtu	Downside / oversupply scenario
$90/t	$3.78/MMBtu	Lower-price case
$100/t	$4.20/MMBtu	Arithmetic anchor used for sensitivities only
$120/t	$5.04/MMBtu	Industrial demand recovery scenario
$140/t	$5.88/MMBtu	Supply disruption / risk premium scenario

Important: no live API2 settlement price is asserted in this paper. The USD 100/t point is an illustrative calculation anchor, and all scenario values are modelling inputs rather than forecasts. ICE API2 futures quote in USD per tonne and are cash-settled against the API 2 index; they are not contracts quoted directly per MMBtu. [4]

04 / ECONOMIC THESIS

Aggregate demand can rise while coal demand falls

The investable question is not whether the economy expands in aggregate, but whether energy-intensive sectors increase their use of the particular coal grades being hedged.



Conceptual, not estimated: stronger demand shifts the demand curve right; lower marginal supply cost shifts supply right. The final price effect depends on the relative size of the shifts and on delivery bottlenecks.

Aggregate demand: channel and limits

• Potential demand support: stronger industrial output, infrastructure investment and demand for steel, cement and chemicals may raise some coal-related industrial inputs. Growth in imports can also increase the value of hedging imported grades and associated logistics.

• Why the link is weaker in Britain: coal-fired electricity generation ended in 2024; steel production processes are changing; the last coke oven at Port Talbot closed in March 2024; and 2025 domestic coal demand was already exceptionally low. [1]

• Macro versus commodity demand: aggregate demand (C + I + G + X - M) is economy-wide spending, not a coal-demand equation. A rise in aggregate demand can be met by gas, nuclear power, renewables, imported electricity, recycled materials or different industrial processes.

Marginal-cost reduction: the test

• Sources of possible cost improvement: productivity, automation, mine planning, energy-efficient ventilation and pumping, better separation / washing, logistics contracts and recovery of usable material from existing sites or spoil where lawful and technically viable.

• Costs that do not vanish: mine safety, water management, geological uncertainty, labour, remediation, restoration, financing, transport, quality penalties, planning and carbon-policy exposure. Full-cycle cost can rise even if a narrow variable-cost measure falls.

• Market-price mechanism: lower marginal costs can shift supply outward and compress benchmark prices if supply expands faster than demand. But a lower-cost British mine does not set the UK or European price unless it meaningfully changes the marginal supply balance.

INVESTMENT RULE

Underwrite a new domestic coal supply proposition only if verified geology, grade, permitted capacity, full-cycle costs and legally accessible buyers support a margin under downside prices. Treat “aggregate demand up” and “marginal cost down” as testable scenarios, not linked certainties.

05 / DERIVATIVES STRATEGY

Trade the liquid benchmark; hedge the local basis

A Barclays programme should begin with established exchange-traded contracts and add a British energy-content index only if the local physical market can support it.

Product layer	Instrument / reference	Commercial use	Key risk
1. Core listed hedge	ICE API2 Rotterdam thermal coal futures and options	European importers, commodity traders and industrial consumers hedge seaborne thermal coal price exposure.	Not a direct hedge for UK-mined anthracite or a different delivery point.
2. Cross-basin hedge	ICE Newcastle or API4/Richards Bay-linked derivatives	Manage export-basin exposure and regional price differentials. [4][5]	Regional basis and freight risk.
3. UK delivered-basis hedge	OTC swap against API2 plus independently evidenced UK freight / quality / port differential	Hedge price paid or received for a specific imported grade delivered into Britain.	Differential data may be thin; contract terms must specify grade and delivery.
4. British Btu-linked pilot	Bilateral price swap on verified £/MMBtu or $/MMBtu domestic anthracite assessment	Could help a qualified domestic producer or buyer secure a grade-adjusted energy price.	Very high index, liquidity, concentration and manipulation risk until a deep cash market exists.
5. Structured protection	Put / call options, collars, caps and floors where permitted	Protect producer floor prices or buyer budgets while shaping upside/downside.	Option premium, volatility, margin and counterparty risks.

How the programme would work

• Producer: hedge a proportion of forecast sale volumes with matched futures / swaps; avoid selling more hedged energy than credible output.

• Buyer: hedge expected purchase volumes and, separately, logistics and quality differentials. Avoid forcing imported API2 exposure onto a different domestic grade without a basis adjustment.

• Barclays intermediary: structure and execute the hedge, offset exposure where possible, price residual basis / liquidity risk transparently and operate within trading limits. The value proposition is risk transfer and intermediation, not a promised directional return.

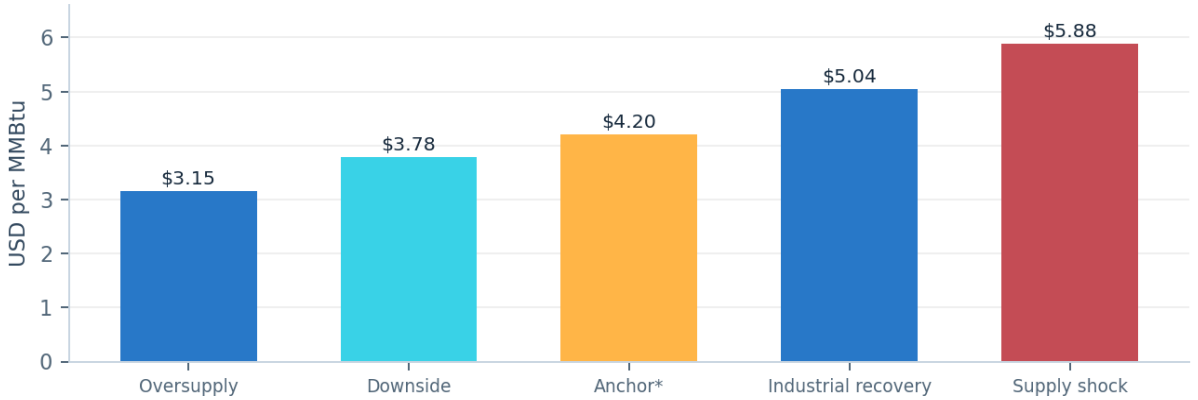
PROPOSED CONTRACT DESIGN - DISCUSSION ONLY

British Coal Energy-Adjusted Basis Swap: reference price = verified domestic coal price per tonne divided by measured MMBtu/tonne, less / plus a documented delivery differential. Settlement would be financial, against an independent published assessment or auditable basket of arm's-length transactions. Do not list the product until data governance, sufficient contributors, minimum transaction volumes, dispute resolution and benchmark manipulation controls are approved.

06 / ILLUSTRATIVE UNIT ECONOMICS

Translate coal-price movements into energy exposure

These examples show how energy-content normalization helps risk managers compare offers and quantify hedge sensitivity. All values are illustrative, not live prices, cost estimates or recommendations.



*Scenario chart uses illustrative coal prices of USD 75-140 per tonne divided by 23.81 MMBtu/tonne. It is a calculation illustration, not a market forecast.

A 1,000-tonne benchmark contract: sensitivity illustration

Metric	Illustrative result	Interpretation
Contract size	1,000 metric tonnes	ICE API2 futures are specified in 1,000-tonne units. [4]
Energy-equivalent quantity	About 23,810 MMBtu	Assumes 6,000 kcal/kg NAR for calculation only. A different grade changes this number.
At USD 100/t	USD 100,000 notional; about USD 4.20/MMBtu	Notional amount before margin, fees and any basis effects.
USD 10/t price move	USD 10,000 per 1,000-tonne contract	Approximately USD 0.42/MMBtu equivalent at reference heat content.
Tick value	USD 50 per contract for a USD 0.05/t tick	ICE API2 tick size is USD 0.05/t and contract size is 1,000 tonnes. [4]

Illustrative marginal-cost sensitivity for a hypothetical 10,000-tonne domestic supply tranche

Assumption	Before productivity gain	After 15% lower variable cost
Variable cost per tonne	£80/t	£68/t
Cost per MMBtu at 23.81 MMBtu/t	£3.36/MMBtu	£2.86/MMBtu
Annual variable-cost saving on 10,000t	-	£120,000 (before extra costs / capex)

The example is arithmetic only. It does not establish that £80/t is a real British mine cost, that a 15% reduction is achievable, or that a new mine is licensable. Investment analysis must add fixed costs, sustaining capital, restoration, transport, quality, financing, taxes and carbon / policy exposure.

07 / IMPLEMENTATION

A staged programme limits liquidity and policy risk

The implementation should separate a viable derivatives desk from a much more speculative domestic mine-development thesis.

Stage	Workstream	Evidence required to proceed	Gate
0-90 days	Market and legal feasibility Map producer / buyer exposures; assess current API2, API4 and Newcastle liquidity; define client eligibility and hedge accounting support.	Client pipeline, exchange-product rules, legal opinions, clear risk ownership and collateral model.	Approve analysis only; no proprietary position assumed.
3-6 months	Benchmark and Btu analytics Build a grade-quality database, cost curves, MMBtu conversion tool, landed-price calculator and AD / supply stress scenarios.	Repeatable data, traceable quality specs, actual delivered prices, validated back-test and governance controls.	Offer analytics and matched hedges against established benchmarks.
6-12 months	Client pilot Run a small number of matched hedges for qualified importers / industrial users; add basis swaps only with documented differential evidence.	Sufficient counterparty limits, stress testing, margin liquidity and measurable client risk reduction.	Scale only if transactions show repeat demand and positive risk-adjusted economics.
12+ months	Domestic Btu index option Assess whether British anthracite / other domestic grades have enough disclosed arm's-length trade to create a reliable index.	Multiple independent contributors, publishable price series, enough volume, robust correction and anti-manipulation framework.	No listed product until the market is demonstrably liquid and legally supportable.

Priority client segments

- Coal importers and traders: seaborne benchmark, freight and quality risk management; likely the most natural first users.
- Steel, cement, lime, chemicals and specialist industry: budget protection for any actual coal, anthracite, coke or related solid-fuel exposure; volumes and fuel substitution must be verified customer by customer.
- Existing domestic coal producers / wholesalers: only where permitted operations, forward output, grade assay and actual sale counterparties can be documented.
- Financing partners: structured working-capital / inventory finance could accompany hedging, but collateral, price volatility, inventory quality and environmental risk should be reflected explicitly.

GO / NO-GO PRINCIPLE

A derivative should be introduced only when a physical customer exposure exists, a sufficiently reliable reference price is available, and Barclays can manage the hedge, basis, liquidity, counterparty and conduct risks under stress. A forecast of industrial growth by itself is not enough.

08 / RISK AND RECOMMENDATION

Do not confuse a useful hedge market with a coal-growth forecast

The risk-managed opportunity is to serve clients exposed to coal prices while preserving a strict hurdle for new British mine supply or a bespoke domestic contract.

Risk	Why it matters	Required control
Policy / licensing	Government intent to restrict future new-mine licences; land-use, planning and climate rules may constrain expansion. [2][3]	Legal review and a permitting milestone before any project-dependent underwriting.
Structural demand decline	UK demand fell 56% in 2025 and coal power demand is zero. [1]	No base case built on a UK power-sector comeback; use sector-specific demand stress cases.
Basis and quality	Domestic anthracite differs from API2 reference thermal coal; heat value, ash, sulphur, moisture, freight and location diverge.	Grade-specific assay and separate basis hedge; no assumed one-for-one hedge correlation.
Thin index / manipulation	Low domestic physical volumes can make a price assessment unrepresentative or vulnerable to concentration.	Independent index governance, multiple contributors, audit trail, fallback / disruption provisions, no listing until robust.
Price / margin / collateral	Short-lived price spikes or rapid falls can generate large mark-to-market losses and cash calls.	Stress testing, position limits, client suitability, liquidity buffers and matched exposures.
Environmental / stranded asset	Carbon pricing, emissions policy, remediation and customer fuel switching can damage asset value and liquidity.	Full-cycle economics, lifecycle emissions, transition scenarios and exit / remediation costs.

Recommendation to the investment committee

1. Proceed: design a coal risk-management proposition around existing liquid seaborne benchmarks, energy-content analytics and matched client hedges.

2. Pilot conditionally: bilateral British delivered-coal or domestic anthracite basis swaps only against auditable transaction data and real client exposures.

3. Defer: any listed “British coal Btu future” and any mine-development investment thesis until market depth, licencing, full-cycle costs and contracted demand meet objective hurdle criteria.

BOTTOM LINE

Rising aggregate demand can lift some industrial coal exposures, and productivity may lower the marginal cost of a lawful existing operation. Neither outcome overturns the present evidence of a small, shrinking UK coal market. The strongest near-term case is a derivatives and client-risk business; a bespoke British Btu index is an option to develop, not the starting assumption.

APPENDIX / SOURCES & ASSUMPTIONS

Source register, formulae and important limitations

Public sources underpin the UK market baseline and exchange-product descriptions. Scenario prices, cost improvements and product designs are expressly illustrative.

[1] Department for Energy Security and Net Zero (DESNZ), DUKES 2026, Chapter 2: Solid Fuels and Derived Gases, published 30 July 2026. UK coal demand, production, imports, industrial use and the end of coal-fired generation. Open source

[2] GOV.UK / DESNZ, New coal mining licences will be banned, 14 November 2024. Government announcement of its intention to legislate to restrict future licences for new coal mines. Open source

[3] Full Fact, Have new coal licences been granted by the government?, updated 6 August 2026. Status tracker noting no licences for new sites since the 2024 election and that the government has said it intends to legislate. Open source

[4] ICE Futures Europe, API2 Rotterdam Coal Futures. Cash settlement against the API 2 Index; 1,000 metric tonnes per contract; USD per tonne quotation; USD 0.05/t minimum price fluctuation. Open source

[5] globalCOAL, Financial Coal Trading. Description of the API2 benchmark for CIF ARA thermal coal at 6,000 kcal/kg NAR and the existence of ICE financial coal benchmarks. Open source

[6] globalCOAL, Thermal Coal Trading. Standardised seaborne thermal-coal products including 6,000 kcal/kg delivered ARA specification. Open source

Technical formulas

Energy content conversion: kcal/kg × approximately 1.8 = Btu/lb. At 6,000 kcal/kg NAR, this is approximately 10,800 Btu/lb; multiplied across a metric tonne it is approximately 23.81 MMBtu/t. Actual as-received coal quality varies; calorific value must be measured and adjusted under the relevant specification.

Price normalization: price per MMBtu = price per tonne ÷ MMBtu per tonne. This arithmetic alone does not adjust for freight, port, ash, sulphur, moisture, grindability, taxes, carbon costs, FX or grade differentials.

Scenario values: USD 75, 90, 100, 120 and 140 per tonne, plus a hypothetical variable-cost reduction from GBP 80/t to GBP 68/t, are illustrative modelling assumptions chosen solely to demonstrate conversion and sensitivity. They are not current coal quotes, forecasts, project cost estimates or commitments.

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